

YASEMIN GOKCEN

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Computational cognitive scientist specializing in human-autonomy teaming and AI systems that understand human language and behavior. Two summers of applied aviation human factors research at NASA Ames and Wisk Aero, and retained by NASA beyond the internship. Trains and evaluates neural language models, links them to human brain data, and builds with AI-assisted development tools (Claude, Codex). Two-time ARCS Foundation Scholar.

CORE COMPETENCIES

AI & Computational Modeling: PyTorch, TensorFlow, HuggingFace Transformers; GPT-2 surprisal and LLM feature extraction; recurrent and working-memory models; AI-assisted development (Claude, Codex)

Statistical & Experimental: mixed-effects modeling (lme4, brms), Bayesian inference; reaction-time and log data cleaning; EEG (EEGLAB, FieldTrip), fMRI (FreeSurfer, SPM), eye-tracking; experimental design, power analysis

Human Factors & Communication: human-autonomy teaming, detect-and-avoid; human-in-the-loop simulation, NASA-TLX workload; research roadmapping, literature synthesis; stakeholder presentation, peer-reviewed publication

Languages & Tools: Python, R, Go, MATLAB, Bash, SQL, Git, SLURM

EXPERIENCE

Research Assistant, Human-Autonomy Teaming Lab | NASA Ames Research Center | Moffett Field, CA | Jun 2026–Present

- Summer internship extended into an ongoing role at the lab's request, continuing human-autonomy teaming research alongside dissertation work
- Owned reaction-time data for a human-in-the-loop study of remote pilots responding to autonomous detect-and-avoid (DAA) alerts; turned unusable raw simulation logs into the analysis-ready dataset used by the whole team
- Designed and ran the reaction-time analysis by alert type and maneuver, began integrating NASA-TLX workload and physiological data, and presented results in a division talk
- Authored a 12-page research roadmap on the changing human role in autonomous aviation: certification challenges, gaps in existing human factors research, and one research area developed into sub-questions, study designs, and data requirements
- Embedded observer for a week-long air traffic control simulation with two retired controllers; produced structured run-by-run records of participant reasoning and traffic responses used by the study team

Human Factors Intern | Wisk Aero (Autonomous Aviation) | Mountain View, CA | Jun–Aug 2025

- Led applied research on human-automation interaction in aviation, defining goals, hypotheses, and success criteria for concurrent initiatives
- Ran rapid prototype testing with eye-tracking, physiological sensors, and qualitative methods; modeled the multimodal data to identify patterns in operator behavior
- Presented findings to senior stakeholders and engineering teams, influencing product decisions, and set the team's post-internship research directions

Graduate Teaching Assistant | University of California, Merced | Aug 2025–May 2026

- Led weekly discussion and computer science lab sections for 100+ students; gave individual feedback on written and coding work

RESEARCH

Context Gating in Language Processing: Computational and Neural Evidence | PhD Dissertation, UC Merced
| 2022–Present

- Developed a computational account of when context management engages during real-time comprehension, the same filtering problem any AI system faces in deciding what to prioritize
- Trained and evaluated 15 recurrent language models (LSTM, RNN variants); LSTMs achieved up to 25% lower surprisal under high memory load, isolating the contribution of context gating
- Collected EEG during naturalistic story listening and modeled N400 and P600 responses with GPT-2 surprisal, semantic similarity, and word frequency as predictors; model residuals correlated significantly with EEG amplitudes
- Built a working memory model in Go (Leabra/Emergent) with SLURM parameter sweeps on an HPC cluster and an R analysis pipeline; ablations across three variants identified the load-bearing memory component
- Published at CogSci 2025 (peer-reviewed, top cognitive science venue); presented at Society for the Neurobiology of Language (2023, 2025); manuscript in preparation (2026)

AI-Assisted Prototyping and Modeling

- Built and deployed a GPT-2 surprisal calculator (JavaScript, GitHub Pages) with Claude and Cursor, concept to live demo in under one hour
- Implemented a predictive coding network from scratch in PyTorch; >90% accuracy on unseen logic gate sequences, outperforming feedforward baselines

AWARDS & LEADERSHIP

- **Graduate Dean's Dissertation Fellowship**, UC Merced (Fall 2026): nomination-only, one award per doctoral program per year
- **ARCS Foundation Scholar** (2024–2026): two consecutive years, one of 80 STEM graduate students across Northern California, nomination-only
- **MacKenzie Scott Travel Award** (2023): international presentation at Society for the Neurobiology of Language, Marseille
- **UC Merced Academic Senate Summer Fellowship** (2023–2024); **ADNiR Scholar**, Ohio State (2021–2022)
- **GradEXCEL Peer Mentor** (2024–Present); **Graduate Student Group Social and Community Officer** (2023–2024)

EDUCATION

PhD, Cognitive and Information Sciences | University of California, Merced | Expected May 2027

Advisors: David C. Noelle (PBWM computational model of working memory) and Rachel Ryskin (language prediction; MIT/Gibson Lab)

MS, Cognitive and Information Sciences | University of California, Merced | May 2026

BS, Cognitive/Computational Neuroscience | The Ohio State University | May 2022